Physics Basis of the Current Reactor Web App Model

1 Purpose of this note

This document explains the physics behind the current simulator implementation in the repository. The model is intentionally simple: it is a *point kinetics* transient model with one control input (control rod insertion), one lumped core, and fixed heavy-water moderation assumptions.

The most important point is the following:

> The current model is **not** a full diffusion or transport calculation of a heavy-water reactor. It is a point-kinetics model whose control-rod worth curve is **inspired by** the fundamental axial shape from one-group diffusion theory.

So the answer to "did you use one-group diffusion?" is:

- **For the transient equations: no.** The time evolution is point kinetics.
- **For the rod worth shape: approximately yes.** The cumulative rod-worth function was chosen so that its *differential* worth follows the kind of $\sin^2$ axial weighting one would expect from the fundamental flux shape in a one-group diffusion picture.

2 What the model assumes

The model makes the standard point-kinetics assumption that the neutron flux can be factored into a fixed shape and a time-dependent amplitude,

$$\phi(\mathbf{r}, E, t) \approx n(t)\, \psi(\mathbf{r}, E),$$

where:

- $\psi(\mathbf{r}, E)$ is a fixed normalized shape,
- $n(t)$ is the time-dependent neutron population amplitude.

This means that all spatial information is collapsed into a single amplitude. In other words, the code assumes that control motion changes reactivity, but does not reshape the flux in space.

That is why the current model can produce:

- reactivity versus time,
- flux amplitude versus time,
- power versus time,

but it cannot yet produce:

- radial power peaking,
- axial power peaking,
- local flux depression around rods,
- explicit moderator or temperature feedback.

3 Point kinetics equations

The transient core model is the standard six-group delayed-neutron point kinetics system:

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^{6} \lambda_i C_i, \tag{1}$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i, \qquad i = 1, \dots, 6. \tag{2}$$

Here:

- $n(t)$ is the normalized neutron population,
- ρ is the reactivity in $\Delta k/k$,
- $\beta = \sum_i \beta_i$ is the effective delayed neutron fraction,
- Λ is the prompt neutron generation time,
- C_i is the precursor concentration of delayed group i,
- λ_i is the decay constant of delayed group i.

The code uses the following values:

$$\beta_{\text{eff}} = 0.00651,$$
$$\Lambda = 5 \times 10^{-4} \text{ s}.$$

The delayed neutron group data currently used are:

Group i	β_i	λ_i [s^{-1}]
1	0.00025	0.0124
2	0.00138	0.0305
3	0.00122	0.1110
4	0.00264	0.3010
5	0.00075	1.1400
6	0.00027	3.0100

These are best interpreted as **generic thermal-reactor point-kinetics data**, not as a lattice-physics fit to a specific heavy-water reactor design.

3.1 Power and flux scaling

The solver evolves only the normalized neutron population. Displayed power and flux are then obtained from

$$P(t) = P_0\, n(t), \tag{3}$$
$$\Phi(t) = \Phi_0\, n(t), \tag{4}$$

with the nominal values

$$P_0 = 250 \text{ MW}_{\text{th}},$$
$$\Phi_0 = 2.4 \times 10^{13} \text{ n}\,\text{cm}^{-2}\,\text{s}^{-1}.$$

So if $n(t)$ doubles, both displayed power and displayed total flux double.

3.2 Numerical time integration

The code does not use an explicit Euler update. Instead it uses a semi-implicit update to avoid instability in this stiff kinetics system.

For each time step Δt, the precursor equations are written as

$$C_i^{k+1} = \frac{C_i^k + \Delta t\,(\beta_i/\Lambda)\,n^{k+1}}{1 + \lambda_i \Delta t}.$$

Substituting these into the neutron balance gives a closed equation for n^{k+1}. This is why the simulator remains stable at the chosen time step even for moderate positive reactivity insertions.

4 Why these reactor dimensions were chosen

The current `REACTOR_MODEL` stores:

$$R_{\text{in}} = 0.8 \text{ m},$$
$$R_{\text{out}} = 2.2 \text{ m},$$
$$H = 5.8 \text{ m}.$$

These define an **annular core** with:

- inner radius $R_{\text{in}} = 0.8$ m,
- outer radius $R_{\text{out}} = 2.2$ m,
- active height $H = 5.8$ m.

The corresponding cross-sectional area is

$$A = \pi\left(R_{\text{out}}^2 - R_{\text{in}}^2\right) = \pi(2.2^2 - 0.8^2) \approx 13.19 \text{ m}^2,$$

and the active volume is

$$V = AH \approx 76.53 \text{ m}^3.$$

At the nominal thermal power $P_0 = 250$ MW$_{\text{th}}$, this corresponds to an average volumetric power density of

$$\frac{P_0}{V} \approx 3.27 \text{ MW}\,\text{m}^{-3}.$$

This is a **reasonable order-of-magnitude engineering choice** for a low-resolution educational model: it gives a core that is neither unrealistically tiny nor excessively dilute.

4.1 What these dimensions do and do not mean

The dimensions were chosen to support a physically sensible interpretation:

- an annular fuel region around a central moderator channel,
- enough core height for a meaningful top-inserted control rod,
- a core volume consistent with the nominal power scale.

However, in the current code these dimensions are **not used to compute**

- critical buckling,
- diffusion coefficients,
- migration area,
- geometric buckling B_g^2,
- leakage,
- rod worth from first principles.

So the geometry is presently **descriptive and interpretive**, not yet predictive. It defines what kind of core the point model is intended to represent, but it is not itself closing the neutronics problem.

5 How the control rod model was constructed

5.1 The implemented reactivity law

Let $x \in [0, 1]$ be the rod insertion fraction, with

$$x = \frac{\text{rod insertion percent}}{100}.$$

The code defines the cumulative rod-worth shape as

$$W(x) = x - \frac{\sin(2\pi x)}{2\pi}.$$

The total rod contribution to reactivity is then

$$\rho_{\text{rod}}(x) = -\rho_{\text{rod,max}}\, W(x),$$

with

$$\rho_{\text{rod,max}} = 700 \text{ pcm}.$$

The total reactivity in pcm is

$$\rho_{\text{tot,pcm}}(x) = \rho_{\text{base}}\rho_{\text{rod}}(x)\rho_{\text{scram}}.$$

where:

- ρ_{base} is chosen so that the reactor is critical at 50% insertion,

- $\rho_{\text{scram}} = -450$ pcm when SCRAM is latched,
- otherwise $\rho_{\text{scram}} = 0$.

Because the critical insertion is set to $x_c = 0.5$,

$$\rho_{\text{base}} = \rho_{\text{rod,max}}\, W(0.5).$$

Since $W(0.5) = 0.5$, this gives

$$\rho_{\text{base}} = 350 \text{ pcm}.$$

Therefore:

- at 50% insertion, the core is nominally critical,
- withdrawing rods below 50% adds positive reactivity,
- inserting rods above 50% adds negative reactivity.

5.2 Why this function is physically plausible

The key observation is that

$$\frac{dW}{dx} = 1 - \cos(2\pi x) = 2\sin^2(\pi x).$$

So the *differential* rod worth is proportional to

$$\sin^2(\pi x).$$

That is exactly the kind of shape one gets if:

1. the core is dominated by the fundamental axial mode,
2. the axial neutron flux is approximated by
$$\phi(z) \propto \sin\left(\frac{\pi z}{H}\right),$$
3. local control rod effectiveness is proportional to the local importance or, in a crude approximation, to $\phi(z)^2$.

Under those assumptions, the differential rod worth is largest near the axial midplane and smallest near the top and bottom ends. Integrating that axial weighting from the top of the core to insertion depth z gives precisely a cumulative function of the form

$$W(x) = \int_0^x 2\sin^2(\pi u)\, du = x - \frac{\sin(2\pi x)}{2\pi}.$$

5.3 So: was one-group diffusion used?

Not as a solved reactor model. The code does *not* solve the one-group diffusion equation

$$-D\nabla^2\phi + \Sigma_a\phi = \frac{1}{k}\nu\Sigma_f\phi$$

for this annular geometry, and it does not compute rod worth by perturbing a diffusion eigenvalue.

But **yes, the rod-worth shape is motivated by the fundamental mode logic** that one would get from one-group diffusion in the axial direction. More precisely:

- the current model *borrows the shape* of a diffusion-theory fundamental mode,
- but it *does not perform* a diffusion-theory criticality or rod-worth calculation.

That makes the current rod model a **physics-inspired ansatz**, not a first-principles diffusion result.

6 Heavy-water interpretation

The code labels the core as heavy-water moderated, but heavy water does not yet appear through explicit moderator equations. In the current model, this means:

- the reactor is intended to be interpreted as a thermalized, D_2O moderated system,
- the annular geometry with a central low-absorption moderator region is meant to be qualitatively consistent with that picture,
- the delayed-neutron and kinetics constants are still generic thermal reactor values rather than a calibrated CANDU-style lattice data set.

So "heavy-water moderated" is presently a **modeling context**, not yet a material-property calculation.

7 What is heuristic and what is derived

7.1 Reasonably physics-based pieces

- Standard six-group point kinetics equations.
- Standard interpretation of reactivity in $\Delta k/k$.
- A rod differential worth proportional to a fundamental-mode $\sin^2$ weighting.
- Power and flux proportional to neutron population amplitude.

7.2 Heuristic or tuned pieces

- The actual annular dimensions are engineering choices, not diffusion-derived.
- The total rod worth of 700 pcm is a tuning choice.
- The extra SCRAM penalty of 450 pcm is a tuning choice to ensure shutdown margin.
- The nominal power and nominal flux are scale-setting choices.
- The model does not yet include temperature, void, xenon, or burnup feedbacks.

8 How to make the model more rigorous

If a more physics-grounded reactor model is desired, the next upgrade would be:

1. Choose a specific reactor type and fuel/moderator composition.
2. Solve a one-group or multi-group diffusion eigenvalue problem in the annular geometry to obtain the spatial flux shape and k_{eff}.
3. Derive geometric buckling from the actual dimensions.
4. Compute control rod worth from perturbation theory or repeated eigenvalue solves with varying insertion depth.
5. Derive β_{eff}, Λ, and nominal flux/power from a consistent lattice/core model.
6. Only then reduce that spatial model to a calibrated point-kinetics surrogate.

9 Bottom line

The current simulator is best described as:

> A point-kinetics reactor transient model with six delayed neutron groups, a heuristically scaled annular heavy-water-reactor geometry, and a control rod worth curve chosen to mimic the axial fundamental-mode weighting suggested by one-group diffusion theory.

That is physically interpretable and useful for an educational transient model, but it is not yet a full spatial reactor calculation.